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(1) Research Program/Project/Study Title: Development of a 5-in-1 Size Reduction Machine	
Proponent Name:	ABELARDO Y. BERMEJO JR
Academic Rank:	Professor
Co-Authors (with academic rank):	Bonifacio Hontiveros (Instructor), Sandra Balsote (Asso Prof), Francis Belle Lampong (Instructor), Kenny John Ballera (Instructor), , Marty Delfin (Instructor), Catherine Arroyo (Asso Prof), Alex Buenafe (Asso Prof), Jonnie Lungsod (Administrative Aide)
Campus/Satellite College/Address:	Pontevedra Campus, Bailan, Pontevedra, Capiz
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation CapSU – Pontevedra Campus Municipality/City: Pontevedra Province: Capiz Region: 6			
(5) Classification Research: ☐ Basic ☐ Applied Development: ☒ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Agricultural crops – coconut, root crops, vegetables		(8) Discipline/Field of Specialization Agricultural and Biosystems Engineering	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement c

Many foodstuffs require preparation or transformation to make them suitable for consumption, or to modify their flavor, texture and digestibility. They undergo either cutting, slicing, chopping, mincing, chipping, grating, or other similar processes. The objective of these processes is also to reduce the size of material either for further processing or to improve the eating quality or suitability of foods for direct consumption.

Cutting is used for the size reduction of large to medium sized parts of food material. There are 14 basic cuts: brunoise (fine dice), chiffonade (shredding), julienne (match sticks cut), macedoine (small dice), parmentier, rondelle (washer), carre, wedges, batons, slicing, crushing, mincing, and paysanne. The food that is **brunoised** includes carrots, onions, leeks, celery, bell peppers, and hard root vegetables like beets and turnips to be used as garnishes and stuffings. **Chiffonade** is a French term that means little ribbons. It is a slicing technique used on leafy green vegetables and herbs such as spinach, sorrel, and flat-leaved herb like basil, mint to be used as ingredient for sautee, stuffing and garnish. **Julienne or allumette** is a French culinary knife cut in which food items are cut into long thin strips. The common vegetable items used for julienne are carrots for carrots julienne, celery for céleris remoulade, potatoes for julienne fries, or cucumbers for naengmyeon. They are also used as a decorative cut to garnish various dishes. **Macedoine** cut is often used in the preparation of soup, stock, or salad which is called Macedoine Salad, which would have the macedoine of vegetables mixed with Mayonnaise. **Parmentier** is a medium-size cube cut. It is the same dice style as to make doing just slightly larger around 1.5 cm for salads and sautee preparation. **Carre** is a large-size cube cut. "**Carré**" means "square" in French. Vegetables like potatoes, and sometimes fruits such as watermelon are cut this way. The **mincing** technique is used where food ingredients are finely divided into uniform pieces. Minced food is normally smaller than dice or chopped. **Mincing** is the ideal technique for aromatics, such as onion, garlic, and ginger, it is used when a paste texture is required for stuffing and garnishing. **Crushing** is used to crush food such as ginger, garlic. This technique is used to turn vegetables into a barrel shape. It is used on carrot, potatoes, or squash that provide a distinctive and consistent to the food items being served. The term **rondel** means round or circular, this technique is used to cut vegetables or fruits into a round shape. foods like cucumber, zucchini, carrots, cucumber, pineapples, strawberry are used. **Paysanne** is a French term that is used to describe vegetables that have been cut into thin slices with shapes that correspond best to the shape of the vegetable. The word **paysanne** means "*country-style*", indicating that this is a rougher, more informal cut than other more precise cuts that exist in French cooking. The paysanne cut ensures a quick and even cooking of vegetable. As an example, carrots may be thinly sliced into circular button shapes or half-round buttons, thus utilizing all of the vegetables as effectively as possible for the food being prepared. **Slicing** is the cutting of food into thin, relatively broad slices. Slices may be used as they are or processed further to produce other speciality cuts such as chiffonade, rondelles, diagonals, oblique or roll cuts, and lozenges. Slicing may be accomplished by hand or machine (Hyfoma, 2018).

Grating, also known as **shredding** is to turn foods into fine pieces. The main difference between shred and grate is that shredding results in longer fragments, while grating results in shorter fragments. Hard vegetables, like carrots, celeriac, beetroot, cauliflower,

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broccoli, peppers and parsnip, are the best suited to this technique. Some vegetables, such as carrots, jicama, turnips, celery root, cucumbers, and zucchini, can be tough to eat in their raw form unless they're cut into small pieces. Shredding them breaks them into softer bits that are easier to eat (<https://weresmartworld.com/culinary-techniques/culinary-technique-grating>). Locally, the **grated** food materials are coconut, cassava, sweet potato, ginger and carrots. The designed root crop grater will help small vendors of traditional food products such as cassava cake, *pitsi-pitsi*, *suman*, *bibingka* or *kakanin*, *camote* delight, *salabat* and also for making a carrot juice.

Stripping of root crops with uniform thickness is essential for all food product processing. The non-uniformity of thickness of the strips may result in uneven absorption of ingredients and other food processing chemicals (Truong, 1985 as cited by Durana et al, 2014). Cassava, sweet potato, potato, banana, bamboo shoot, papaya and others are the ones stripped. Moreover, chipping means to separate by cutting, chopping or breaking. This is suited for root crops or banana which are made into small thin pieces called chips.

The absence of a motorized machine for preparing local foodstuffs in large volume is a problem of food manufacturers. A lot of persons engaged in producing pickled papaya for commercial purposes resort to using improvised shredders made of tin. Makers of *kakanin* also use the traditional method because of the unavailability of an alternative tool to grate their ingredients like cassava and coconut.

There are commercially available machines used to prepare or transform foodstuffs but issues on their use, convenience, cost and safety. Most of them can perform only one or few functions. Some are done manually, hence the quality of foodstuff and efficiency are low and the operator tends to exert more time and effort. Moreover, owning a very costly processing machine is not feasible. There are also electronic and manually-operated machines but one has to watch his fingers, as some of them have parts that are quite sharp.

Due to the foregoing reasons and unsatisfactory conditions in the preparation of foodstuffs, developing a machine which can do various functions is thought of. Hence, this study “Development of a Multifunction Processing Machine” is conducted.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

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Figure 1. The proposed multifunction size-reduction machine

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(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study generally aims to fabricate a multi-function processing machine. Specifically, it aims to:

- (1) Design a multi-function processing machine.;
- (2) Fabricate the designed multi-function processing machine.;
- (3) Evaluate the fabricated multi-function processing machine. in terms of:
 - (a) Cutting capacity
 - (b) Shredding capacity
 - (c) Slicing capacity
 - (d) Grating capacity
 - (e) Sizes of the processed food stuff
- (4) Perform cost analysis of using the developed multi-function processing machine.
- (5) Compare the capacity and products of the newly designed multi-function processing machine with the manual method.

(11) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

The machine will work by doing size reduction of agricultural crops either cutting, slicing, chopping, mincing, chipping, grating, or other similar processes.

The machine's designs and methods of tests will be guided by the Philippine Agricultural Engineering Standards (PAES).

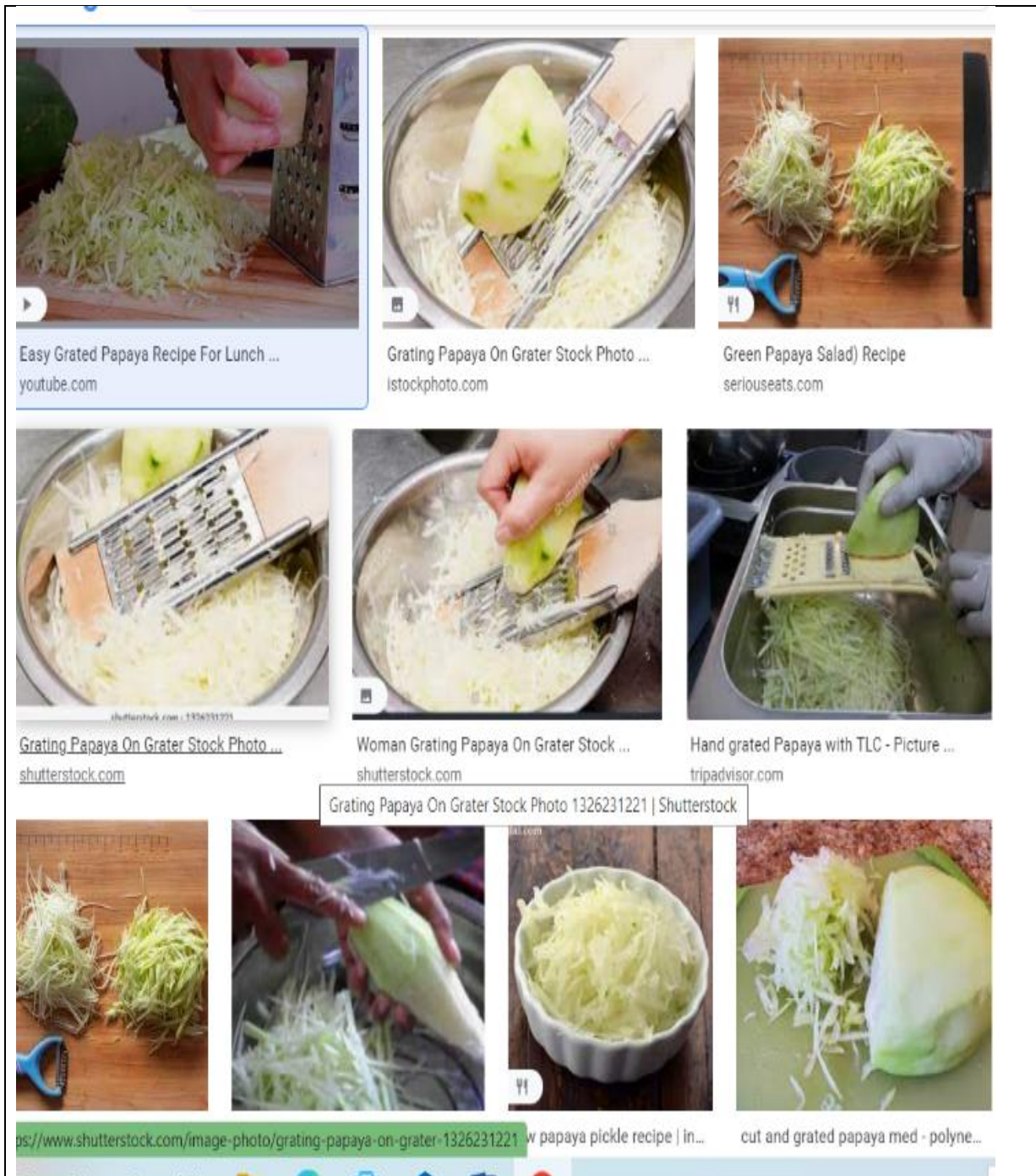
(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

B02C18/062 - Disintegrating by knives or other cutting or tearing members which chop material into fragments with rotating knives with rotor elements extending axially in close radial proximity of a concentrically arranged slotted or perforated ring

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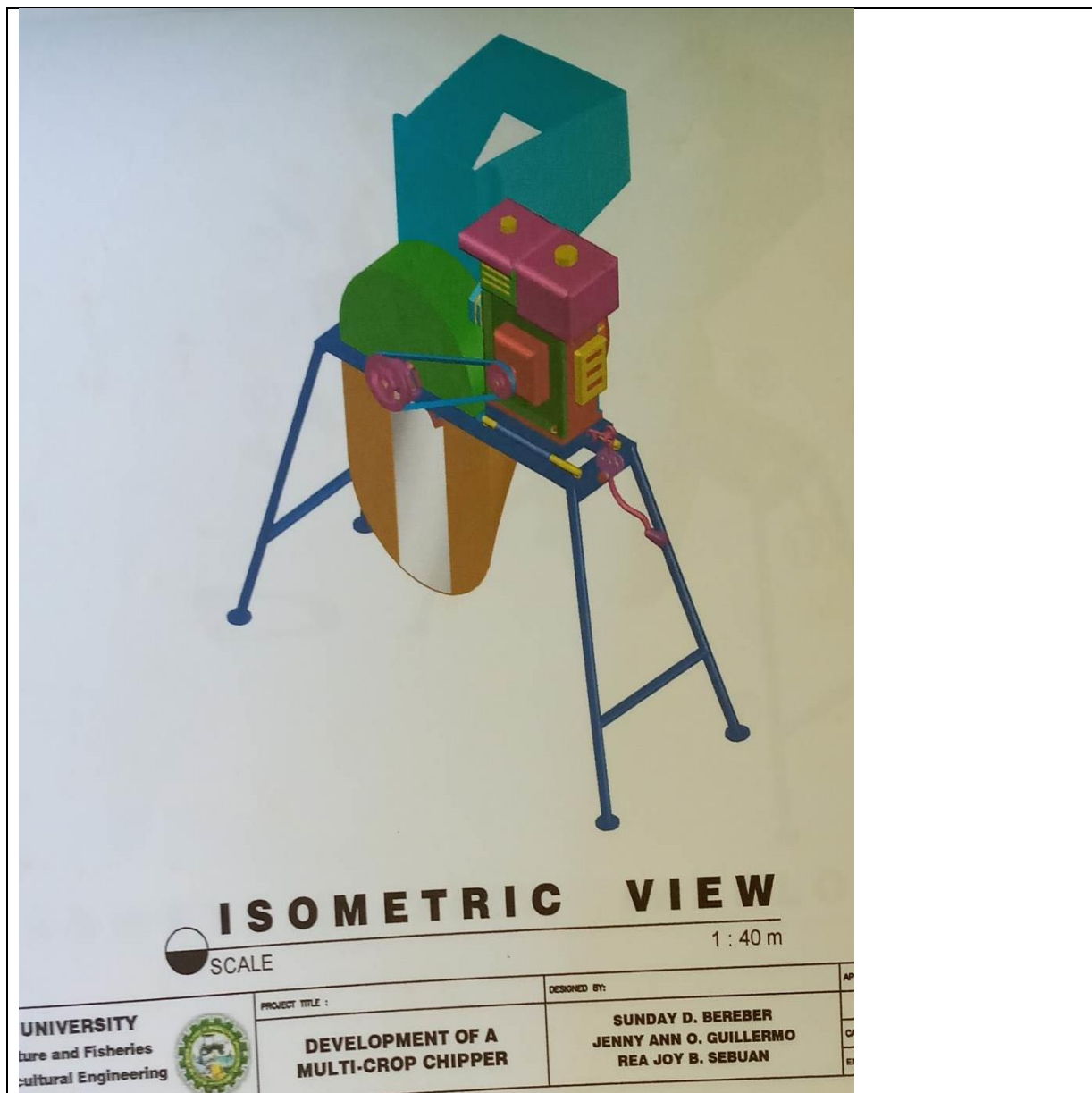
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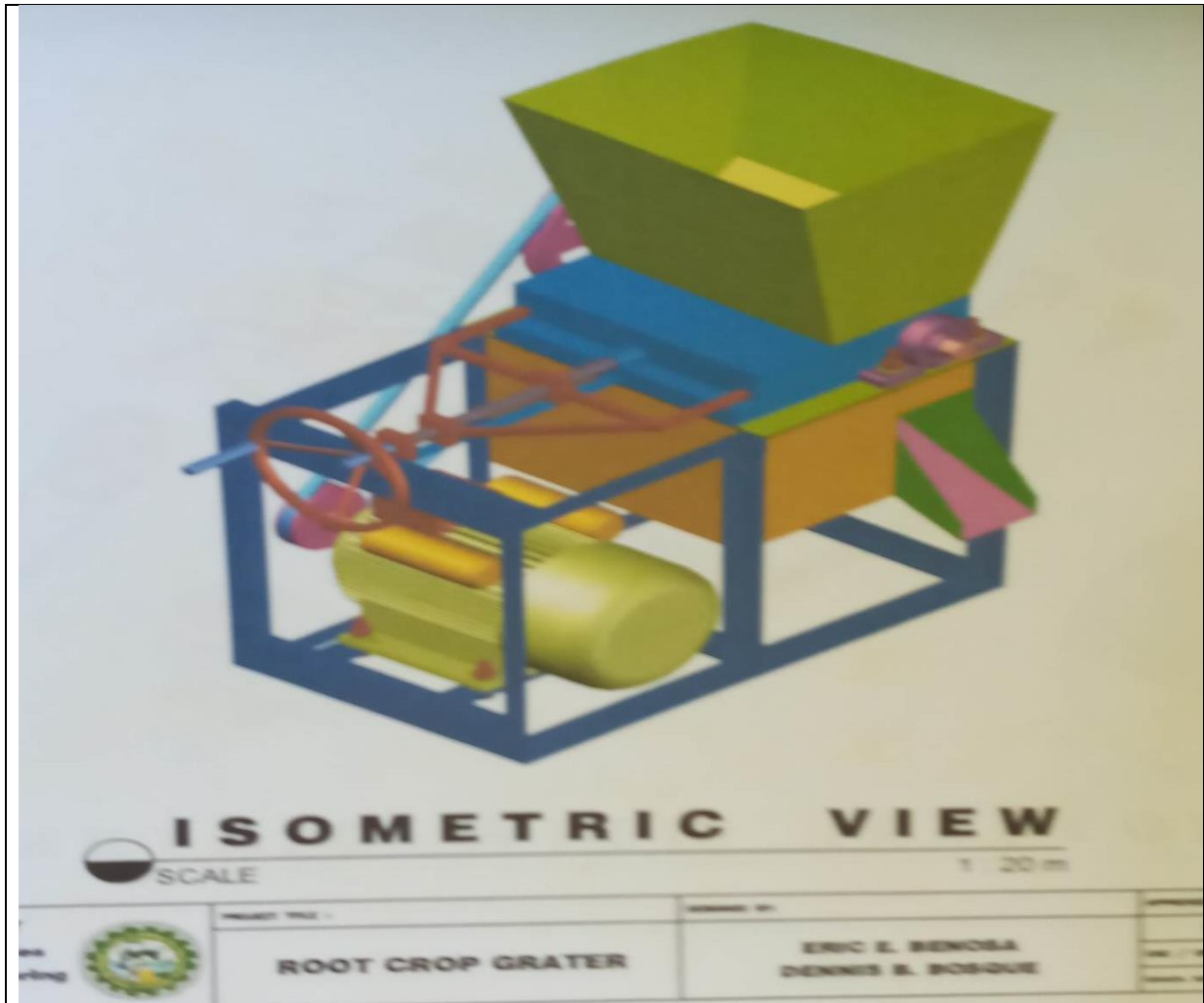
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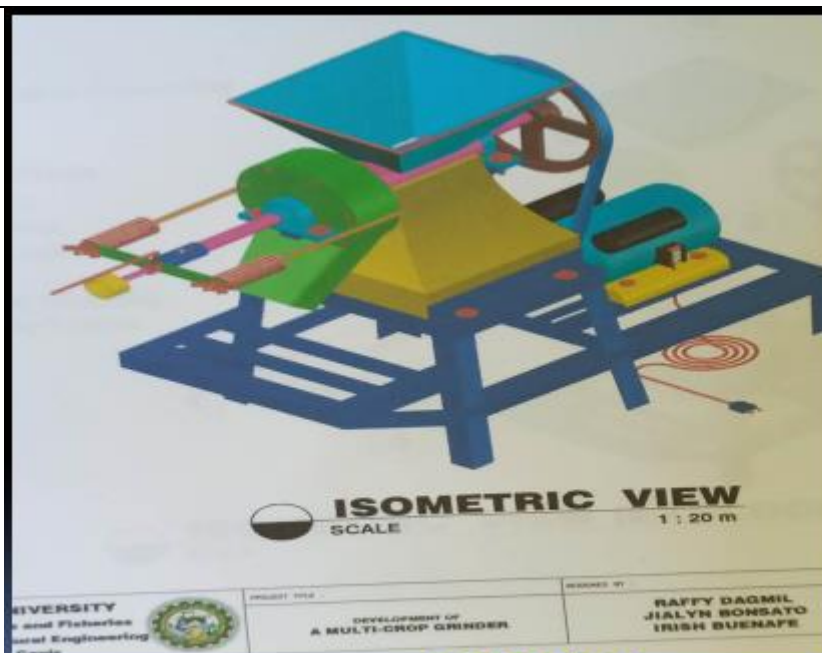
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PRIOR-ART



PRIOR-ART

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

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Designing the 5-in-1 Size Reduction Machine

The design of the developed multifunction processing machine will be based from the design of existing coconut graters of Capapas (2005) and Mamburam et al (2016) as well as the root crop grater of Benosa and Bosque (2015). It was also be based from the improvised coconut grater seen in the locality.

The criteria to be considered in designing the multifunction processing machine are as follows: (a) ease of loading, (b) ease of cleaning the parts, (c) ease of adjustment and repair of parts, (d) ease of collecting output, (e) ease of transporting the machine, (f) safety, (g) vibration/noise level, (h) quality of output or product..

Fabrication of the 5-in-1 Size Reduction Machine

The materials needed in the study will secured through purchase order with the supply office with the support of the Research Department . The machine will be fabricated at the shop of CapSU Pontevedra. One of the authors will be requested to be relieved from his work to concentrate on the fabrication of the machine.

Tha fabricator will be provided with a sketch of the design. The fabrication will go through the following processes: lay-outing, cutting, welding, boring, riveting, bending, bending and sanding.

Evaluation Procedure of the 5-in-1 Size Reduction

The fabricated machine will be evaluated in terms of (a) cutting, grating, slicing and shredding output capacity (g) cost and (h) design criteria.

The output capacity will be determined by getting the time spent in cutting, grating, slicing and shredding of the foodstuffs as well as the weight of the foodstuff.

The cost of the machine will be expressed in Philippine peso. It will be determined by adding up the cost of materials purchased and used in the fabrication of the machine. The feasibility of using it will consider the following: fixed and variable costs, usage per day and per week. Fixed costs include depreciation, interest on investment, taxes, insurance and shelter (TIS) and repair and maintenance. Variable costs include labor and electric power consumption.

The evaluation criteria will be rated by professors, non-academic personnel, agricultural engineering students and graduates and community members. It used a five-point

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scale as follows: 5 (Very good), 4 (good), 3 (satisfactory), 2 (poor) and 1 (very poor). The data will be analyzed using means and standard deviation.

The cost of the machine will be expressed in Philippine peso. It will be determined by adding up the cost of materials purchased and used in the fabrication of the machine. The feasibility of using it will consider the following: fixed and variable costs, usage per day and per week. Fixed costs will include depreciation, interest on investment, taxes, insurance and shelter (TIS) and repair and maintenance. Variable costs will include labor and electric power consumption.

The evaluation criteria will be rated by professors, non-academic personnel, agricultural engineering students and graduates and community members. It will use a five-point scale as follows: 5 (Very good), 4 (good), 3 (satisfactory), 2 (poor) and 1 (very poor). The data will be analyzed using means, standard deviation, t-test and analysis of variance.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified based on your desired Work plan.)

SCHEDULE OF ACTIVITIES

ACTIVITY	JANUARY 2023	FEBRUARY 2023	MARCH 2022	APRIL 2023	MAY 2023	JUNE 2023
Purchase of Materials						
Fabrication						
Testing&Evaluation						
Manuscript Writing						

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The expected outputs of this research are (a) a design drawn using AUTOCAD, (b) fabricated size-reduction machine, (c) data taken from the testing and evaluation, (d) manuscript

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The research output will be (a) subjected to technology protection, (b) demonstration or adoption by target beneficiaries/ community, (c) brought to exhibits/fairs, (d) subjected to patent/utility model application

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(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Coconut farmers, business individuals, food processors and institutions engaged in agricultural crops

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Abelardo Y. Bermejo	Project Leader	Agricultural Engineer
Bonifacio A. Hontiveros	Co-author/Member	Mechanical Engineer
Sandrfa S. Balsote	Co-author/Member	Agriultural Engineer
Kenny John P. Ballera	Co-author/Member	Agricultural Engineer
Francis Belle D. Lampong	Co-author/Member	Agricultural Engineer
Catherine B. Arroyo	Co-author/Member	Civil Engineer
Marty C. Delfin	Co-author/Member	Civil Engineer
Alex D. Buenafe	Co-author/Member	Electrical Engineer
Jonnie Lungsod	Co-author/Member	TechVoc (Structural Welding and Automotive Mechanic)

Percent time devoted to the Program/ Project/ Study: 20%

Program/Project Leader (in hours): 8 hours/week

Program/Project Members (in hours): 8 hours/week

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 research asst for 3 months xP13,000/month	P39,000.00
B. Maintenance and Other Operating Expenses (MOOE)		
- Transportation allowance		5,000.00
- Meal allowance		5,000.00
- Professional fees (laboratory analysis and other related fees – fabricator, machining, etc)		20,000.00
- Miscellaneous (for incidental expenses – coconut, papaya, carrots, patato, etc..)		20,000.00
- Printing (bond paper, printer, ink)		15,000.00
C. CAPITAL OUTLAY (construction materials and Research instruments (weiging scale, tachometer, noise level meter, stop watch, caliper)		50,000.00
Total		P154,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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BENOSA, E. and D. BOSQUE. 2016. Design, Fabrication and Evaluation of a Root Crop Chipper. Undergraduate Thesis. Capiz State University, Pontevedra Campus. Agricultural Engineering Department.

CAPAPAS, C. 2005. Design, Fabrication and Evaluation of a Manual-Operated Coconut Grater. Undergraduate Thesis. Capiz State University, Pontevedra Campus. Agricultural Engineering Department.

DURANA, C., A. DIACONTE and A. Depio. 2014. Design, Fabrication and Evaluation of a Crop Chipper-Dicer. Undergraduate Thesis. Capiz State University, Pontevedra Campus. Agricultural Engineering Department.

MAMBURAM, L. and P. DAYAL. 2016. Design, Fabrication and Evaluation of a Coconut Grater. Undergraduate Thesis. Capiz State University, Pontevedra Campus. Agricultural Engineering Department.

RUIZ, J. and L. FINEZA. 2017. Design, Fabrication and Evaluation of a Vegetable Slicer. Undergraduate Thesis. Capiz State University, Pontevedra Campus. Agricultural Engineering Department.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

ABELARDO Y. BERMEJO, JR.

Licensed Agricultural Engineer

Professor

BSAE, MAT Mathematics, Ed.D.

25 years teaching experience

Researches Conducted: Development of (a) tractor-mounted lawn mower, (b) young coconut fruit trimmer, (c) corn sheller, (d) rice harvester, (f) plow

BONIFACIO A. HONTIVEROS

Licensed Mechanical Engineer

Instructor

BSME, MA Mathematics

14 years teaching experience

Researches Conducted: Development of (a) tractor-mounted lawn mower, (b) young coconut fruit trimmer, (c) rice mill, (d) rice thresher, (e) paddy collector, (f) shell washer

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SANDRA S. BALSOTE

Licensed Agricultural Engineer

Instructor

BSAE, MA Mathematics

30 years teaching experience

Researches Conducted: Development of (a) young coconut fruit trimmer, (b) briquetting machine, (c) fish dryer with gasifier (d) turmeric juice extractor

KENNY JOHN P. BALLERA

Licensed Agricultural Engineer

Instructor

BSAE, MS Agronomy

8 years teaching experience

Researches Conducted: Development of (a) tractor-mounted lawn mower, (b) young coconut fruit trimmer, (c) corn sheller, (d) rice thresher, (e) root crop chipper

FRANCIS BELLE D. LAMPONG

Licensed Agricultural Engineer

Instructor

BSAE, MS Agricultural Engineering

1 year teaching experience

Researches Conducted: Development of (a) biogas digester (b) coconut fruit trimmer with gearbox system

CATHERINE B. ARROYO

Licensed Civil Engineer

Asso Prof

BSCE, MAEd

28 years teaching experience

Researches Conducted: Development of (a) young coconut fruit trimmer (b) coconut fruit trimmer with gearbox system, (c) smokehouse stove, corn mill

MARTY C. DELFIN

Licensed Civil Engineer

Instructor

BSCE, MS Math

15 years teaching experience

Researches Conducted: Development of (a) young coconut fruit trimmer (b) coconut fruit trimmer with gearbox system, (c) coconut dehusker

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ALEX D. BUENAFE

Licensed Electrical Engineer

Asso Prof

BSEE, MA Math

28 years teaching experience

Researches Conducted: Development of (a) young coconut fruit trimmer (b) coconut fruit trimmer with gearbox system

JONNIE LUNGSOD

Administrative Aide IV (Electrician)

Tech Voc Course: Structural Welding, Diesel Automotive Mechanic

TESDA NC Holder

10 years experience

Research Conducted: (a) Development of young coconut fruit trimmer, (b) coconut fruit trimmer with gearbox system